

TW13: Applied Data Pre-processing Case Study

Five challenges | 20 marks each | 100 marks total

Student name	
Student ID	
Class / group	
Submission date	

Weekly purpose

Complete an end-to-end retail case study from business question and import through cleaning, EDA, feature engineering, and model-ready preprocessing.

SUBMISSION RULES

- Complete work in the corresponding Jupyter notebook.
- Write your own code and explanations; no solutions are supplied.
- Keep supplied data unchanged and work on copies.
- Restart the kernel and run all cells before submission.
- Submit the notebook with outputs visible and assistance acknowledged.

MARKING RUBRIC FOR EACH CHALLENGE

Assessment area	Marks
Correct method and working code	8
Validation and quality checks	4
Explanation and interpretation	4
Reproducibility and presentation	4
Total per challenge	20

Challenge 1: Define the case-study contract

Scenario

A retail manager wants to know which regions and channels show strong revenue with acceptable returns.

INSTRUCTIONS

1. State the decision, audience, unit of analysis, scope, primary measures, and exclusions.
2. Import the dirty sales file and record provenance.
3. Validate the expected schema and identify unexpected columns.
4. List at least four limitations before analysis.

REQUIRED SUBMISSION

- A complete project contract.
- A provenance record and schema comparison.
- A clearly bounded business question.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 2: Build a reproducible cleaning function

Scenario

The project must rerun from the untouched source with no hidden manual edits.

INSTRUCTIONS

1. Write a function that copies input, removes exact duplicates, standardises labels, parses dates, masks invalid ranges, treats missingness, and validates output.
2. Return both the clean table and a decision log.
3. Compare before-and-after rows, missing cells, duplicates, and invalid values.
4. Use at least eight assertions.

REQUIRED SUBMISSION

- A reusable function with no mutation of raw.
- A decision log and before-after scorecard.
- A validated clean DataFrame.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 3: Perform structured exploratory analysis

Scenario

The manager needs evidence on scale, transaction value, return rate, satisfaction, and time.

INSTRUCTIONS

1. Calculate overall orders, customers, revenue, median transaction, return rate, and mean satisfaction.
2. Create region-by-channel and category summaries.
3. Plot one group comparison and one monthly trend.
4. Identify one trade-off and perform one sensitivity check using a robust statistic.

REQUIRED SUBMISSION

- Overall and grouped summary tables.
- Two purposeful charts.
- A finding, sensitivity result, and limitation.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 4: Engineer model-ready features safely

Scenario

The analysis may later predict returns at order placement.

INSTRUCTIONS

1. Create revenue and calendar features using only information available at prediction time.
2. Create a role table identifying IDs, predictors, target, post-outcome fields, and exclusions.
3. Make a stratified train/test split before learned preprocessing.
4. Build numeric and categorical ColumnTransformer pipelines.
5. Fit on training only and verify equal output widths with finite values.

REQUIRED SUBMISSION

- An engineered feature table and role audit.
- A leakage-safe split and fitted preprocessor.
- At least five transformed-matrix assertions.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 5: Deliver the case-study handover

Scenario

Another analyst must be able to understand, rerun, and challenge the complete case study.

INSTRUCTIONS

1. Create a feature manifest and data-quality scorecard.
2. Write an executive conclusion with finding, evidence, limitation, recommendation, and next validation.
3. Create a reproducibility checklist covering paths, seeds, environment, raw data, and run order.
4. Rerun from a clean kernel and record the result.

REQUIRED SUBMISSION

- Feature manifest, scorecard, and reproducibility checklist.
- A 200-to-250-word executive conclusion.
- A signed-off handover record.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Final submission checklist

- ☐ All five challenges are complete.
- ☐ The notebook runs from top to bottom without errors.
- ☐ Every output is clearly labelled.
- ☐ Important results include written explanations.
- ☐ Validation checks are included.
- ☐ Supplied raw data has not been overwritten.
- ☐ Sources and assistance are acknowledged.

Student declaration

I confirm that this submission is my work, that I have acknowledged sources and assistance, and that the displayed outputs were produced by the submitted code.

Signature: _____ Date: _____